

Distributed Linux File Sharing System

1. Introduction

This project was done to build and test a small distributed file sharing system using Oracle VM VirtualBox, Ubuntu Desktop and Ubuntu Server. Two virtual machines were connected using a VirtualBox host-only network. Static IP addresses were configured so that the two machines could communicate with each other. SSH was then used for remote access and SCP was used to transfer files between the machines.

2. Project Objectives

- Create and configure two Ubuntu virtual machines in Oracle VM VirtualBox.
- Connect the two virtual machines to the same virtual network.
- Configure static IP addresses and test communication between the machines.
- Use SSH and SCP to demonstrate remote access and file sharing.
- Demonstrate common Linux commands and record their results.
- Simulate a network failure, observe its effect, and restore communication.

3. Virtual Machine Setup

Two virtual machines were created in Oracle VM VirtualBox. One machine was installed with Ubuntu Desktop and the second machine was installed with Ubuntu Server. The machines were kept on the same VirtualBox host-only network so they could communicate directly.

VM 1 – Ubuntu Desktop

IP address: 192.168.56.101/24

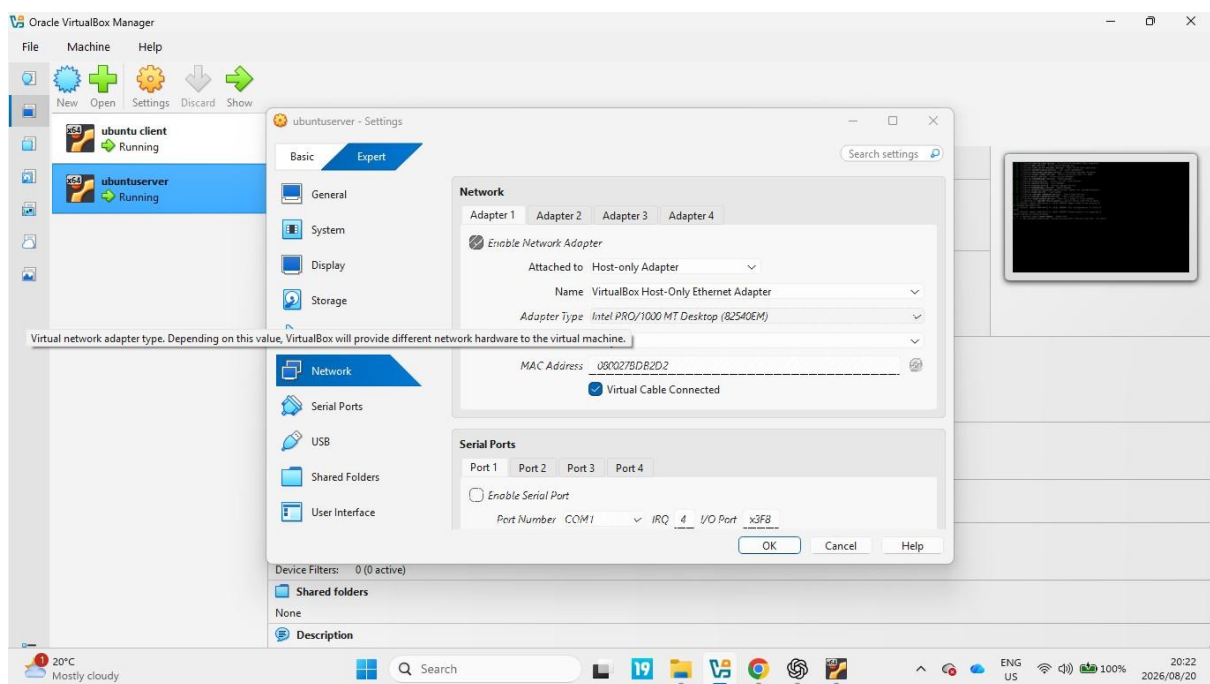
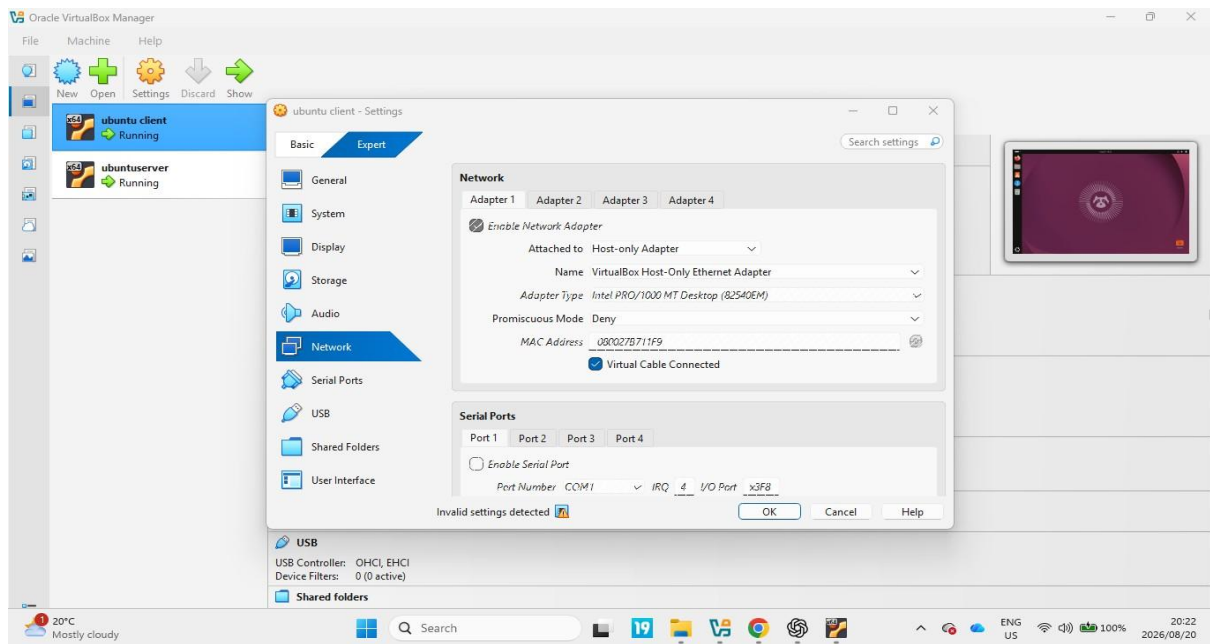
Interface: enp0s3

VM 2 – Ubuntu Server

IP address: 192.168.56.102/24

Interface: enp0s3

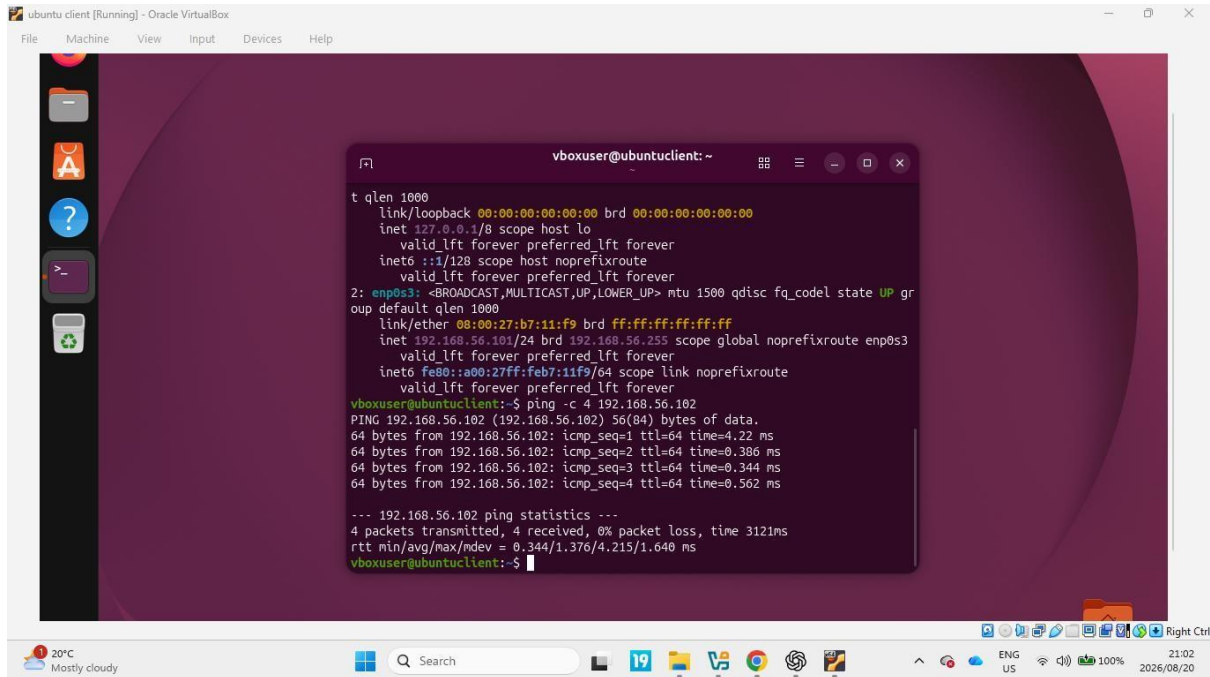
The subnet used was 192.168.56.0/24, with 255.255.255.0 as the subnet mask.



4. Network Configuration and Connectivity

The two virtual machines were connected to the same host-only network. Static IP addresses were then configured. The Desktop was assigned 192.168.56.101/24 and the Server was assigned 192.168.56.102/24.

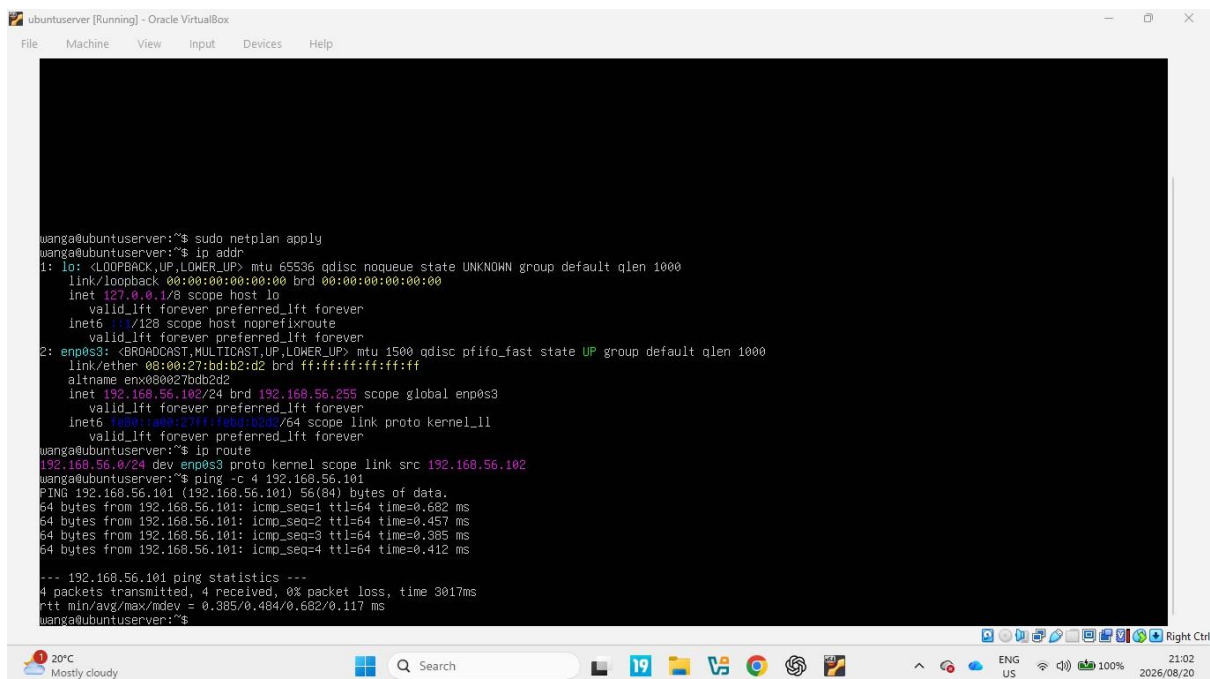
Connectivity was tested in both directions using ping. The Desktop successfully reached the Server and the Server successfully reached the Desktop. The successful tests showed that the two machines could communicate across the virtual network.



The screenshot shows an Ubuntu Desktop environment with a terminal window open. The terminal displays the output of the `ip netns exec` command, showing the network configuration for the `vboxuser` namespace. The configuration includes a loopback interface `lo` and an ethernet interface `enp0s3` connected to the host's `192.168.56.101`. The terminal also shows the output of the `ping` command, indicating a successful connection to `192.168.56.102`.

```
t q len 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
link/ether 08:00:27:b7:11:f9 brd ff:ff:ff:ff:ff:ff
inet 192.168.56.101/24 brd 192.168.56.255 scope global noprefixroute enp0s3
    valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:feb7:11f9/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
vboxuser@ubuntuclient:~$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=4.22 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.386 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.344 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.562 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3121ms
rtt min/avg/max/mdev = 0.344/1.376/4.215/1.640 ms
vboxuser@ubuntuclient:~$
```



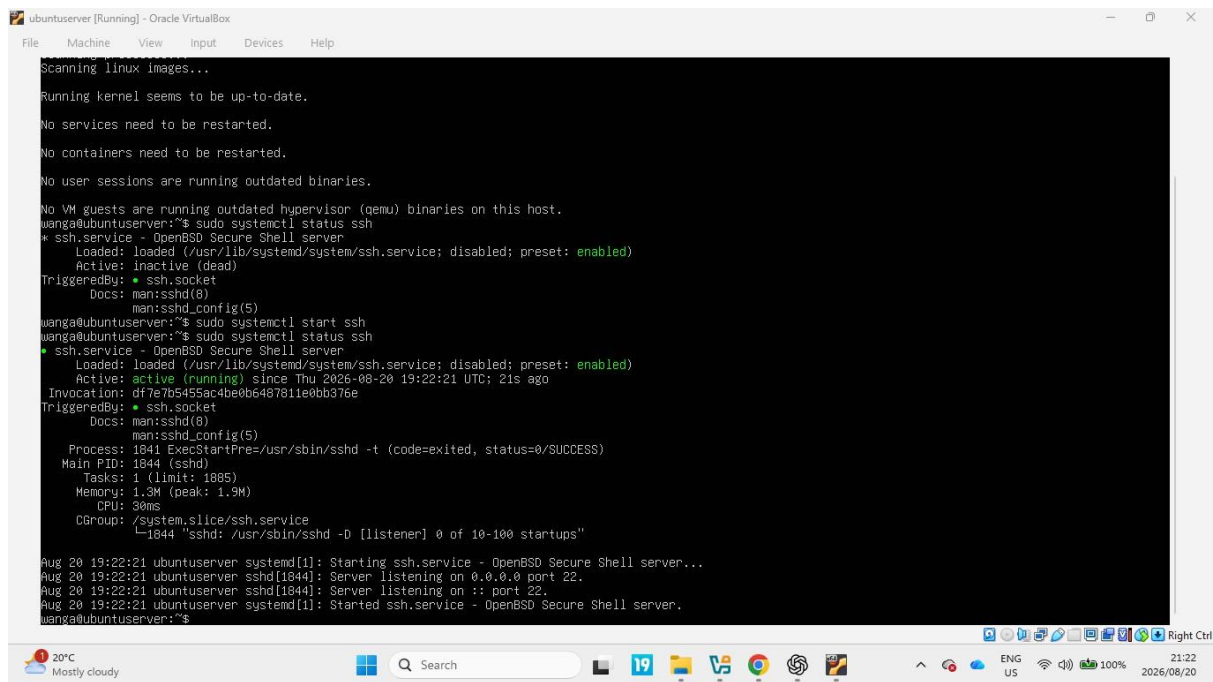
The screenshot shows an Ubuntu Server environment with a terminal window open. The terminal displays the output of the `ip netns exec` command, showing the network configuration for the `wanga` namespace. The configuration includes a loopback interface `lo` and an ethernet interface `enp0s3` connected to the host's `192.168.56.101`. The terminal also shows the output of the `ping` command, indicating a successful connection to `192.168.56.102`.

```
wanga@ubuntuclient:~$ ip netns exec
wanga@ubuntuclient:~$ ip netns exec
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:bd:b2:d2 brd ff:ff:ff:ff:ff:ff
altname enx080027bdb2d2
inet 192.168.56.102/24 brd 192.168.56.255 scope global enp0s3
    valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:feb7:11f9/64 scope link proto kernel ll
    valid_lft forever preferred_lft forever
wanga@ubuntuclient:~$ ip route
192.168.56.0/24 dev enp0s3 proto kernel scope link src 192.168.56.102
wanga@ubuntuclient:~$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.682 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.457 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.385 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.412 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3017ms
rtt min/avg/max/mdev = 0.385/0.484/0.682/0.117 ms
wanga@ubuntuclient:~$
```

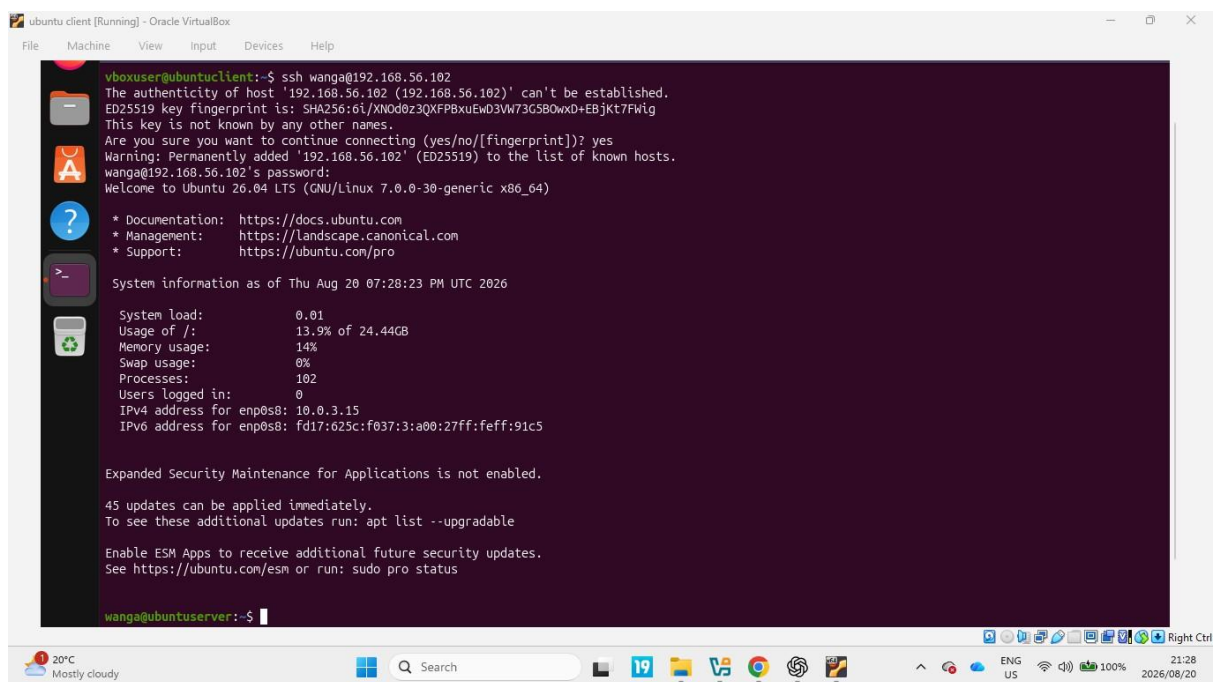
5. SSH Remote Access

OpenSSH Server was installed on Ubuntu Server. The SSH service was started and verified as active and running. From Ubuntu Desktop, an SSH connection was made to `192.168.56.102` using the Server account. This confirmed that the Desktop could access the Server remotely.



```
Scanning linux images...
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
wanga@ubuntu server:~$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
wanga@ubuntu server:~$ sudo systemctl start ssh
wanga@ubuntu server:~$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-08-20 19:22:21 UTC; 21s ago
   Invocation: df7e7b5455ac4be0b6487811e0bb376e
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 1841 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1844 (sshd)
   Tasks: 1 (limit: 1885)
   Memory: 1.3M (peak: 1.9M)
   CPU: 39ms
   CGroup: /system.slice/ssh.service
           └─1844 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 20 19:22:21 ubuntu server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 20 19:22:21 ubuntu server sshd[1844]: Server listening on 0.0.0.0 port 22.
Aug 20 19:22:21 ubuntu server sshd[1844]: Server listening on :: port 22.
Aug 20 19:22:21 ubuntu server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
wanga@ubuntu server:~$
```



```
vboxuser@ubuntucient:~$ ssh wanga@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is: SHA256:6i/XN0d0z3QXFPBxuEwD3VW73G5B0wXD+EBjKt7FWig
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
wanga@192.168.56.102's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Aug 20 07:28:23 PM UTC 2026

System load:          0.01
Usage of /:           13.9% of 24.44GB
Memory usage:         14%
Swap usage:           0%
Processes:            102
Users logged in:      0
IPv4 address for enp0s8: 10.0.3.15
IPv6 address for enp0s8: fd17:625c:f037:3:a00:27ff:feff:91c5

Expanded Security Maintenance for Applications is not enabled.

45 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

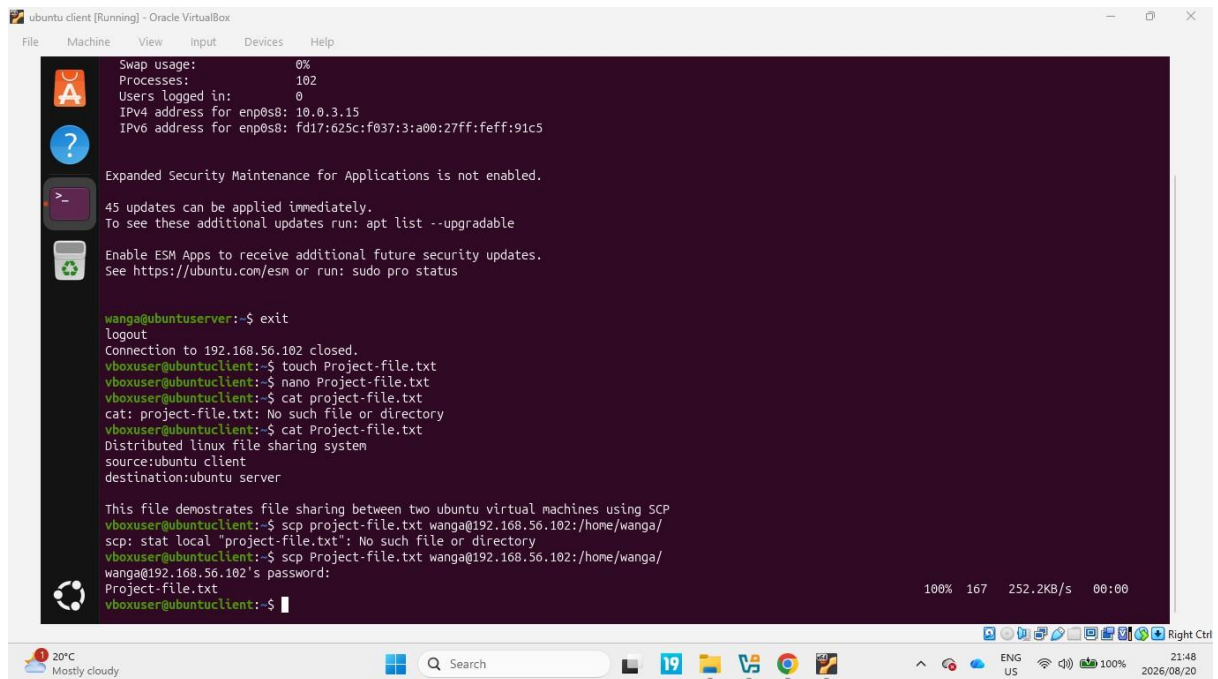
wanga@ubuntu server:~$
```

6. File Sharing Using SCP

SCP (Secure Copy Protocol) was used to transfer files between the two Ubuntu machines. A project file was created on Ubuntu Desktop and copied to Ubuntu Server. The file was then checked on the Server to confirm that the transfer was successful.

A second file was created on Ubuntu Server and copied back to Ubuntu Desktop using SCP. The file was opened on the Desktop after the transfer to confirm that the contents were received correctly.

Transfer 1: Ubuntu Desktop → Ubuntu Server



```
ubuntu client [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Swap usage: 0%
Processes: 102
Users logged in: 0
IPv4 address for enp0s8: 10.0.3.15
IPv6 address for enp0s8: fd17:625c:f037:3:a00:27ff:feff:91c5

Expanded Security Maintenance for Applications is not enabled.

45 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

wanga@ubuntuclient:~$ exit
logout
Connection to 192.168.56.102 closed.
vboxuser@ubuntuclient:~$ touch Project-file.txt
vboxuser@ubuntuclient:~$ nano Project-file.txt
vboxuser@ubuntuclient:~$ cat Project-file.txt
cat: Project-file.txt: No such file or directory
vboxuser@ubuntuclient:~$ cat Project-file.txt
Distributed linux file sharing system
source:ubuntu client
destination:ubuntu server

This file demonstrates file sharing between two ubuntu virtual machines using SCP
vboxuser@ubuntuclient:~$ scp Project-file.txt wanga@192.168.56.102:/home/wanga/
scp: stat local "Project-file.txt": No such file or directory
vboxuser@ubuntuclient:~$ scp Project-file.txt wanga@192.168.56.102:/home/wanga/
wanga@192.168.56.102's password:
Project-file.txt
100% 167 252.2KB/s 00:00
vboxuser@ubuntuclient:~$
```

Transfer 2: Ubuntu Server → Ubuntu Desktop

ubuntu client [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
vboxuser@ubuntuclient: ~  
vboxuser@ubuntuclient:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr  
    link/ether 08:00:27:b7:11:f9 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global noprefixroute enp0s3  
        valid_lft forever preferred_lft forever  
    inet6 fe80::a00:27ff:feb7:11f9/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr  
    link/ether 08:00:27:00:0b:cf brd ff:ff:ff:ff:ff:ff  
    altname enx080027000bcbf  
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 82138sec preferred_lft 82138sec  
    inet6 fd17:625c:f037:3:447f:b076:e137:d2a5/64 scope global temporary dynamic  
        valid_lft 86299sec preferred_lft 14299sec
```

Home

19°C Mostly cloudy 23:32 2026/08/20

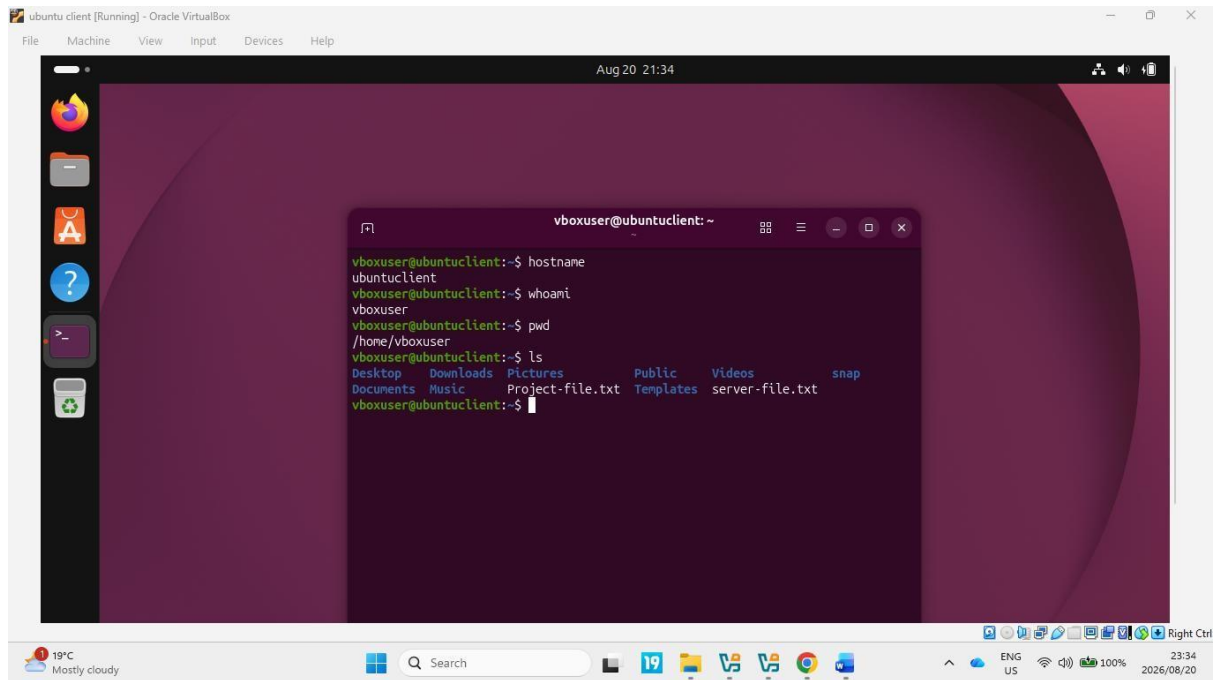
ubuntu client [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Aug 20 21:33

```
vboxuser@ubuntuclient: ~  
vboxuser@ubuntuclient:~$ ip route  
default via 10.0.3.2 dev enp0s8 proto dhcp src 10.0.3.15 metric 101  
10.0.3.0/24 dev enp0s8 proto kernel scope link src 10.0.3.15 metric 101  
192.168.56.0/24 dev enp0s3 proto kernel scope link src 192.168.56.101 metric 100  
vboxuser@ubuntuclient:~$
```

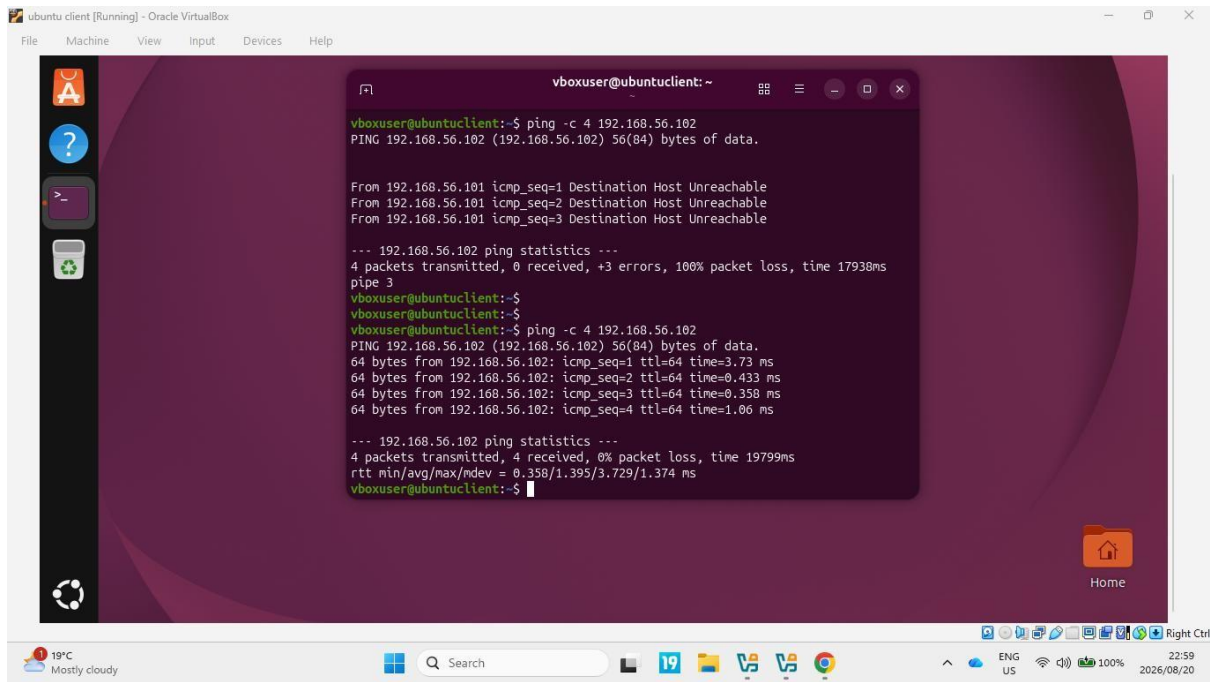
23:33 2026/08/20



8. Fault Simulation and Recovery

A network failure was deliberately simulated on the Ubuntu Server by disabling the `enp0s3` network interface. The Desktop was then used to ping the Server. The ping failed and packets were lost because the Server's network interface was no longer available for communication.

The problem was recovered by bringing the `enp0s3` interface back up. After the interface was restored, the Desktop was able to ping 192.168.56.102 successfully again. This demonstrated both the effect of a network failure and the recovery process.



9. Results

The project requirements were tested practically. Both Ubuntu virtual machines were created and connected to the same virtual network. Static addressing and two-way communication were confirmed. SSH remote access worked, and SCP successfully transferred files in both directions. A network failure was also simulated and communication was restored after the interface was enabled again.

10. Problems Encountered and Solutions

One issue occurred when OpenSSH could not initially be installed because the Server did not have Internet access through the host-only network. A second VirtualBox adapter using NAT was added to provide Internet access while keeping the host-only adapter for communication between the two project machines. After this, the Ubuntu packages could be updated and OpenSSH Server was installed.

Another practical issue was the amount of RAM available on the physical computer. The virtual machines were using too much memory, which made the host computer slow. The VM memory allocation was reduced so that the project could continue without overloading the host computer.

11. Conclusion

The Distributed Linux File Sharing System was successfully implemented using Ubuntu Desktop, Ubuntu Server and Oracle VM VirtualBox. The two virtual machines communicated using static IP addresses, and SSH provided remote access to the Server. SCP was used to transfer files in both directions, demonstrating practical resource sharing. The fault simulation

also showed how loss of a network interface affects communication and how the connection can be restored. Overall, the project provided practical experience with Linux networking, remote administration and secure file transfer.

12. Evidence Checklist

- VirtualBox with Ubuntu Desktop and Ubuntu Server
- Desktop static IP 192.168.56.101/24
- Server static IP 192.168.56.102/24
- Successful ping between both machines
- Hostname/IP verification
- SSH service active and running
- Successful SSH connection
- SCP Desktop → Server
- SCP Server → Desktop
- Linux command screenshots
- Failed ping during simulated fault
- Successful ping after recovery